GZero - User Manual

Break-junction analysis software. Version 1.5, October 2026

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1. Introduction and installation

I wrote GZero to analyse single-molecule break-junction data (MCBJ and STM-BJ) in one place, from the raw TDMS files of the setup to the figures for a paper. It converts the raw signals to conductance, cuts out the single traces, makes the usual histograms, measures plateau lengths, calculates 2D correlation maps, clusters traces and gets the flicker-noise exponent. It also tells you how often a molecule actually bridged the gap (the junction yield) and whether the conductance drifted during the run. For I-V sweeps it fits the single-level model and finds the transition voltage. For each breaking trace it can find the moment the junction ruptures and the steps between conductance levels (change-point detection). For conductance-time traces it finds flickering and mechanical switching events, and for fixed-gap I(t) measurements the molecules blinking in and out of the gap. A separate tab does the ensemble statistics of large-area (EGaIn) junctions, and another one the thermopower (Seebeck coefficient) from thermovoltage measurements. Piezo modulation is covered too, and there is a simulator that makes fake data with known answers, which I use to check settings.

1.1 The tabs

Tab	What it's for
Convert TDMS	Raw TDMS channels to conductance, `.cvr` export
Data & Traces	Loading files, finding and cutting traces, looking through them, marking bad ones
Histograms	1D log G histogram, 2D conductance-distance histogram, Gaussian fits
Plateau length	How long the molecular plateau is, in up to three windows; tunnelling decay
Junction statistics	Which traces have a molecular plateau (junction yield), the conductance of each plateau, drifts over the measurement
Change points	Rupture of every trace (Chow test) and its plateau conductance; the steps between conductance levels (PELT)
Correlation	2D cross-correlation of conductance values within traces
Clustering	PCA and clustering of traces into groups
Flicker noise	Noise power against conductance and the scaling exponent n
Events	Flickering and mechanical switching in G(t), conductance-time histograms of the holds
Blinking	Fixed-gap I(t): molecules jumping in and out of the gap, their conductance, lifetime and arrival rate
I-V	Forward and backward bias sweeps, dI/dV, 2D I-V histograms, single-level model fit, transition voltage spectroscopy
Piezo modulation	Decay constant β from a modulated piezo
EGaIn (ensemble)	Large-area junctions: log|J| statistics, yield, rectification, V_{trans}, β from a length series
Thermopower	Thermovoltage histograms per temperature difference, Seebeck coefficient, S against conductance
Simulation	Synthetic recordings

1.2 What you need

Windows 10 or 11 (64 bit) for the exe. From Python it should also run on macOS and Linux, though it has mainly been tested on Windows. 8 GB of RAM is fine, also for big files. TDMS files above 300 MB are read in pieces (4.7): a 1.3 GB test file (90 min at 10 kHz) loads in about 4 s and needs well under 1 GB of memory. Text files are still loaded completely, so very large text files need about 3-4 GB of RAM per GB of file. The window needs about 1400 x 900 pixels to be comfortable.

1.3 Installing and starting

The simplest way is the exe. Copy the whole `GZero` folder somewhere, e.g. `C:\Programs\GZero`, and double-click `GZero.exe`. Don't move the exe out of the folder, it needs the files next to it. This manual is in `docs\User_Manual.pdf` and also opens from Help -> User manual (F1).

The source code is on GitHub: https://github.com/arman-rashid/GZero (Apache License 2.0). To run it from Python (3.9 or newer), go to the project folder (the one with `GZero.py`) and run

```
py -m pip install -r requirements.txt
py GZero.py
```

It needs PySide6, numpy, scipy, matplotlib, pandas, scikit-learn, npTDMS, igor2 (for Igor `.pxp` / `.ibw` files), openpyxl (Excel) and h5py (HDF5 and MATLAB v7.3 files).

If you change the code and want a new exe: `py -m pip install pyinstaller`, then `py build_exe.py` in the same folder. The new version ends up in `dist\GZero\`.

1.4 The window

Every tab has the settings and buttons on the left and the plots on the right. The toolbar above each plot lets you zoom, pan and save what's on screen. When you save results, each plot is also written as a separate figure in journal format (13.1). Most settings have a tooltip, so hover over them if something isn't clear. The status bar at the bottom says what the program is doing; longer calculations run in the background and the window doesn't freeze.

Settings are only kept while the program is open. File -> Save settings writes them to a file and File -> Load settings reads them back. You can also load the `*_config.json` from an old results folder, which is the easiest way to repeat an analysis with exactly the same settings.

2. Quick start

This is the short version of a normal analysis, from a raw file to a histogram and a noise exponent. The chapter numbers point to the details.

1. Go to Data & Traces, click Load files and pick a `.tdms`, `.cvr`, text, Excel, MATLAB, HDF5, NumPy, LabVIEW `.lvm` or Igor `.pxp` file (several at once is fine, or use Load folder; 3.1 lists all formats). For everything except TDMS check the column settings first: which column is what, and the bias if the file has no voltage column (4.2).
2. The traces are found automatically after the first load. Under the button you see how many were found, how many were accepted and why the rest were thrown out.
3. Click through a few traces with $<$ and $>$ or the arrow keys. Green means accepted, red rejected. If good-looking traces are being rejected, change the limits (4.4) and press Detect / cut traces again.
4. G and B mark the current trace as good or bad by hand.
5. Open Histograms. The 1D and 2D histograms are already there.
6. Put *fit from* and *fit to* around the molecular peak, choose how many Gaussians, click Fit peaks (5).
7. In Plateau length, set window 1 around the peak and click Calculate (6).
8. In Flicker noise, click the Morris et al. 2025 preset, then Peak from histogram fit, then Run noise analysis. n and its error appear under the button (9).
9. Correlation and Clustering if you need them (7, 8). Junction statistics gives the yield and checks for drift (6.4), Change points the conductance of each junction from its rupture (6.5).
10. File -> Save results (Ctrl+S), choose where and give the folder a name. Everything goes in there: text files, figures, settings (13).

If you just want to try the program, go to Simulation and click Generate. You get a fake data set where you know the right answers, and you can see if the analysis finds them (12).

3. Data formats and the Convert tab

You only need the Convert tab for raw `.tdms` files. It turns them into conductance (G/G_0) and either writes `.cvr` files or loads the result straight into the analysis. All other file types are loaded in Data & Traces (4).

3.1 File types

Format	What's in it	Where to load it
`.tdms`, raw	Group `BJRawData` with the bias (`M_E#` or `M_E1-E2`), low-gain current `M_i#a`, high-gain current `M_i#b` and piezo `U_Piezo`, plus the `Setup`, `Calibration_M#` and `Events` groups	Convert tab or Data & Traces
`.tdms`, with G	A `G` channel in siemens and `U_Piezo` (the `*_M.tdms` files)	Data & Traces
`.tdms`, other	Any channel names: type them into the channel settings (3.3)	Convert tab or Data & Traces
`.cvr`	Two tab-separated columns, back-calculated current (A) and piezo (nm). Old files with three columns work too	Data & Traces
`.txt .csv .dat .tsv`	Any number columns; you say which column is what	Data & Traces
`.pxp` (Igor packed experiment), `.ibw` (Igor binary wave)	Every numeric wave is a column, named after the wave (`folder:wave` inside data folders, `wave[0]`, `wave[1]` ... for 2D waves)	Data & Traces
`.lvm` (LabVIEW measurement file)	Tab-separated text with a LabVIEW header. The columns get the names from the header; the `X_Value` column becomes `time`, and `Delta_X` sets the sampling rate	Data & Traces
`.xlsx`, `.xlsm` (Excel)	Every sheet is read. The row above the first all-numeric row gives the column names; columns of the second and later sheets are called `sheet:column`	Data & Traces
`.mat` (MATLAB)	Every numeric variable is a column (`var.field` for struct fields, `var[0]`, `var[1]` ... for matrices). v7.3 files are HDF5 and are read with h5py. A scalar variable `fs`, `Fs`, `sampling_rate` or `dt` sets the sampling rate	Data & Traces
`.npy`, `.npz` (NumPy)	`.npy`: one array, a matrix gives the columns 0, 1, ...; `.npz`: every array by its name, a scalar `fs` or `dt` sets the sampling rate	Data & Traces
`.h5`, `.hdf5` (HDF5)	Every numeric data set is a column, named by its path (`group:dataset`); an attribute `fs`, `sampling_rate` or `dt` sets the sampling rate	Data & Traces
`.txt.gz`, `.csv.gz`, ...	Any of the text formats compressed with gzip; read directly	Data & Traces

All formats except TDMS end up as the same thing inside the program: a list of named columns. So the column settings (4.2) work the same way for all of them, and Show columns (4.1) lists what a file contains. Matrices are always turned so that the longer side is time.

TDMS files carry the sampling rate, channel gains, piezo scale and series resistor. Text files don't, so for those you type them into the column settings (4.2). Igor waves carry their x scaling: if a wave has x in seconds, the sampling rate comes from it.

The program never recalculates what a file already gives. If there is a conductance column or channel, G is taken from it. If there is a voltage column or bias channel, it is used as the bias and the constant bias is ignored. Piezo, current and time columns are also used as they are. Only what's missing is calculated: G from the current and the bias, or the current from G and the bias.

The current can come from a single-stage amplifier (one current, Ia) or a dual-stage amplifier (a low-gain stage Ia and a high-gain stage Ib recorded at the same time). The *current amplifier* setting says which one (3.3 for TDMS, 4.2 for the other files). With a single stage there is nothing to merge and all the Ib settings are greyed out.

3.2 Using the tab

1. Add files or Add folder. When you click a file, the panel shows its sampling rate, length, modules, channel gains, series resistor and whether diode calibration data is in it.
2. Change the conversion settings if needed (3.3). Empty fields mean "take it from the file".
3. Preview selected shows the first 2 minutes of the file: the raw current channels, log(G/G_0) with the event marks (green = approach, red = retract, grey = hold) and the piezo. Selecting a file only reads its header, so this is quick for files of any size.
4. Then either Convert all -> .cvr, which writes one `.cvr` per TDMS file (tick Also write .csv files if you also want a table with time, G, log G, piezo, bias and current), or Load all into analysis, which converts in memory and jumps to Data & Traces. The `.cvr` and `.csv` files are written 2 minutes at a time, so the file size doesn't matter. Each piece is converted with a few thousand samples of overlap on both sides, so the filters give exactly what a conversion of the whole file gives. Load all into analysis reads big files in pieces too (4.7).

3.3 Conversion settings

Setting	Meaning	Default
module	Which measurement module to use (`M_E#`, `M_i#a`, `M_i#b`)	1
current amplifier	`dual stage`: Ia (low gain) and Ib (high gain) are merged. `single stage`: only the Ia channel is used, nothing is merged	dual stage
Ia channel / Ib channel	`auto` = `M_i<module>a` / `M_i<module>b`. Type another channel name if your file uses other names; a channel that is already in A needs gain 1	auto
conductance channel	`auto` uses a `G` channel only if there are no current channels. Type a channel name to take G directly from that channel, `none` to always calculate it	auto
conductance unit	Unit of the conductance channel: `auto` (from the channel, otherwise S), `S`, `G/G0` or `log G/G0`	auto
piezo channel	`auto` = `U_Piezo`; another name, or `none`	auto
calibration	`linear`: I = (V - offset) x gain x correction. `diode`: the piecewise polynomial stored in `Calibration_M#` (log amplifier)	linear
gain Ia / gain Ib (A/V)	Amplifier gains; empty = the `scale_factor` stored with the channel	from file
offset / correction Ia, Ib	Offset (V) and correction factor for the linear calibration	0 / 1
amplification	Extra factor on the current	1
invert current	Flips the sign	off
combine Ia / Ib	Dual stage only. `auto` uses the high-gain Ib unless it's saturated, then Ia. `a` or `b` uses only one	auto
Ib saturation (V)	Above this, Ib counts as saturated	10
Ia saturation (V)	Above this, the contact conducts more than the amplifier can measure and the sample gets the saturated G	10.4
switch to Ib below log G	Optional: only use Ib below this log(G/G_0)	off
bias / bias channel	`measured` takes the bias channel as an array (`auto` = `M_E1-E2` if it exists, otherwise `M_E<module>`, or type a name) and the fixed bias isn't used; `fixed` uses the fixed bias. Without a bias channel the fixed bias is used	measured, auto

Setting	Meaning	Default
series R compensation / series R	Corrects for the current-limiting resistor, G = I / (E - I x R). Empty R = value from the Setup group	on, from file
saturated G (G_0)	What saturated samples are set to	100
piezo scale (nm/V)	Piezo voltage to elongation; empty = the channel's `scale_factor`	from file
bias / current / conductance filter	none, Savitzky-Golay, median or moving average. The window is 2 x side points + 1. Can work on log values	bias: moving average, 25 points

3.4 What happens in the conversion

The current is calibrated and, for a dual-stage amplifier, the two gain channels are put together (Ib where it isn't saturated, Ia elsewhere). The conductance is the current divided by the voltage that's actually across the junction, i.e. the bias minus what drops over the series resistor:

$$G = \frac{I}{E - IR} \cdot \frac{1}{G_0}, \qquad G_0 = \frac{2e^2}{h} = 7.748 \times 10^{-5}\ \mathrm{S}$$

If the file has a conductance channel and you choose it, none of this is done: G comes straight from the channel and only the piezo and bias are read. If the resistor takes the whole bias (E - I x R <= 0), or the low-gain amplifier saturates, the junction is in contact and beyond what can be measured, so the sample is set to the saturated G.

The `.cvr` file contains the back-calculated current, I_bc = G x G_0 x E, which is the current you'd get without the series resistor. When a `.cvr` is loaded, G is calculated back as I_bc / bias / G_0, so the bias in the column settings has to be right.

About the diode calibration: `Calibration_M#` holds five polynomials (PH, PL, M, NL, NH, coefficients in ascending order), an offset and two thresholds. With u = V - offset, the middle polynomial gives I directly when |u| is below the low threshold; outside that, the low or high polynomial gives log10 |I|, and negative u uses the negative-side polynomials with a minus sign. I haven't checked this against a measurement with module 2 yet, so test it with a known resistor before trusting it.

4. Data & Traces tab

Here the recordings are loaded and cut into single breaking traces (making traces too, if you want them). All the other tabs only see the traces that are accepted here.

4.1 Loading

Load files takes one or several files, Load folder takes every file of a readable type (3.1) in a folder. Everything you load is analysed together. TDMS files are converted with the settings from the Convert tab (3.3), text files with the column settings below. Remove drops the selected recordings and their traces, Clear drops everything. Show columns lists the columns (or Igor waves) of a file with their numbers and lengths, so you know what to type into the column settings. The tooltip of a loaded recording says which quantities came from the file and which were calculated. The list shows how long each recording is and whether it has a piezo signal; click one to see it in the overview plot.

4.2 Column settings (every format except TDMS)

A column is given by its number (0 is the first) or by its name: the header of a text file or the wave name in an Igor file. Empty means the file doesn't have it.

Setting	Meaning	Default
conductance column	G given directly. If set, G is read from it and not calculated	empty
conductance unit	`G/G0`, `S` or `log G/G0`	G/G_0
current amplifier	`single stage`: one current column. `dual stage`: Ia and Ib, merged	single stage
current Ia column	The current (single stage) or the low-gain stage Ia (dual stage)	0
current Ib column	The high-gain stage Ib (dual stage only)	empty
Ia scale / Ib scale (A per unit)	Turns the column into A: 1 if it is in A, the amplifier gain in A/V if it is in V	1
Ib saturated from |Ib|	Column value from which Ib counts as saturated and Ia is used instead; empty = 99.5 % of the largest |Ib| in the file	empty
Ia saturated from |Ia|	Column value from which the contact is beyond the range, G = saturated G	off
use Ib below log G	Optional: only use Ib below this log(G/G_0)	off
voltage column	Measured bias in V. Replaces the constant bias, and I-V analysis needs it	empty
constant bias (V)	Only used without a voltage column. For `.cvr` files this must be the bias of the measurement	0.1
series R (ohm)	Series resistor, G = I / (V - I x R) / G_0	0
saturated G (G_0)	What saturated samples are set to	100
piezo / distance column	Piezo position or distance; empty if there is none (then distance = point number x piezo rate, 4.5)	1
piezo scale (to nm)	Factor that turns the piezo column into nm	1
time column (s)	If there is one, the sampling rate is taken from it	empty
sampling rate (Hz)	Used when neither a time column nor the Igor x scaling gives it	10000

What is calculated depends on what you give:

Columns given	Conductance	Current
conductance (+ current)	from the file	from the file, or G x G_0 x V
Ia (single stage)	I / (V - I R) / G_0	Ia
Ia and Ib (dual stage)	from the merged current	Ib where not saturated, Ia elsewhere

V is the voltage column if there is one, otherwise the constant bias.

Files with one wave or column per trace (common in Igor): put a `*` in the conductance or Ia column, e.g. `G_*`. Every matching column becomes its own recording. Other columns with a `*` are paired by the same text, so with piezo column `Z_*` the wave `G_12` gets `Z_12`. Columns of different length (Igor waves) are cut to the shortest one.

Header lines are skipped, and tab, comma, semicolon or space as separator all work.

4.3 How the traces are found

With a piezo signal (`segmentation = piezo`) the piezo is smoothed and its speed calculated. Each point is then either moving forward, moving back, or held, which means the speed is below *hold if speed below (fraction)* times the typical speed. Very short pieces (below *min segment*) are merged

with their neighbours. Every moving piece is a candidate trace and every held piece is a hold.

A moving piece counts as breaking if the conductance drops by more than a decade from start to end, and as making if it rises. If you'd rather decide by piezo direction, set *breaking direction* to `piezo+` or `piezo-`.

A lot of measurements stop the piezo in the middle of a pull to record G(t) for the noise analysis. With *join pull-hold-pull* switched on (default) the pulls before and after the stop are joined into one trace, but the hold points themselves are left out, otherwise the time spent standing still would show up as a fake peak in the histogram. The hold is kept separately for the noise tab (9).

Without a piezo signal (`segmentation = threshold`) the contacts are found where log G goes above the high limit, and each gap between two contacts is split at its lowest point into a breaking and a making trace.

4.4 Cutting limits and checks

All limits are log10(G/G_0). Making traces are turned around and cut the same way, so every trace goes from contact to tunnelling.

Setting	Meaning	Default
high limit to check	The trace has to get above this (metal contact), otherwise it's rejected	0
low limit to check	The trace has to get below this, otherwise it's rejected	-4.5
start at	The cut trace starts at the last point above this; empty = the high limit	empty
low limit to cut	The trace ends at the first point below this after the contact ...	-5.0
additional length	... plus this many points	100
reject noisy start	Throws out traces whose first *noise points* scatter more than *noise limit* (std of log G)	off, 25, 0.2
max floor noise	Optional, throws out traces whose last points are noisier than this	off
use traces	`break`, `make` or `both`	break

I usually put the low limits about half a decade above the noise floor. You can see the floor in the overview plot as the flat band at the bottom.

4.5 Distance axis

Setting	Meaning	Default
distance from	`piezo` uses the smoothed piezo signal, `index` uses point number x piezo rate	piezo
displacement ratio	Junction displacement per nm of piezo (the MCBJ attenuation factor; 1 for STM)	1
piezo rate	nm per point, for `index` and for files without piezo	0.001
zero set	Distance zero is where the trace first drops below this log G (-0.3 is 0.5 G_0)	-0.3
piezo smoothing	Smoothing window for direction detection and the distance axis	10 ms

The piezo has to be smoothed because in my files its noise (about 0.003 nm per sample) is as big as the step between samples. For an MCBJ the displacement ratio can be calibrated with the tunnelling

fit in the Plateau length tab (6.3).

4.6 Overview and trace browser

The overview plot shows log G against time for the selected recording, with the piezo on the right axis. Breaking traces are shaded red, making traces blue and holds grey; light shading means rejected.

Below it, the trace browser shows one trace against distance with its own histogram next to it. The line above the plot gives the trace number, type, file, length and whether it's accepted (and if not, why).

Key / button	What it does
Left / Right arrow, < >	Previous / next trace
G, Good	Mark as good and go to the next one
B, Bad	Mark as bad and go to the next one
Auto	Go back to the automatic decision
Accepted only	Only browse accepted traces
Overlay	Plot N random accepted traces on top of each other

Manual marks are kept until you detect the traces again.

4.7 Large TDMS files

A 1 GB TDMS file holds tens of millions of samples per channel, and converting it in one go needs several GB of memory, because every step of the conversion makes another array as long as the file. That is what made long recordings crash or crawl. So TDMS files above *read in pieces above* are never loaded at once.

When you load such a file, the program goes through it one piece at a time (*piece length*, 2 minutes by default), converts each piece with the normal settings and only keeps a min/max overview of log G and the piezo for the plot. The file list says "read in pieces" next to it.

Detect / cut traces then goes through the file a second time. Each piece is read together with a *margin* (60 s) on both sides, cut into traces with the same code as always, and only the traces and holds that start inside the piece itself are kept. A trace that crosses the border between two pieces is therefore found exactly once, and in full. After that, only the traces, the holds and the overview are in memory, and that is all the Histograms, Plateau length, Junction statistics, Correlation, Clustering, Flicker noise and Events tabs need.

Two things follow from this. If you change the cutting settings and detect again, the file is read again, so don't move or delete it while you're working. And the I-V and Piezo modulation tabs need the complete signals, so for a file read in pieces they read only the part between *from* and *to* (in s) in their Recording box, or the whole file if you leave these empty.

Setting	Meaning	Default
read in pieces above (MB)	TDMS files bigger than this are read in pieces	300
piece length (s)	Length of one piece; about 100 MB of memory per minute of 10 kHz data	120
margin (s)	Read before and after every piece; must be longer than one trace (pull, hold and pull)	60
overview points	Points of the overview plot	400000

Reading in pieces doesn't change the result. On a simulated recording it gave exactly the same traces and holds as loading the whole file (the same sample ranges, log G, distance axes and accept/reject

decisions), even with pieces as short as 20 s, and the self-test checks this every time it runs. On a 1.29 GB file (54 million samples per channel, 90 minutes at 10 kHz, 1760 traces and 3520 holds) loading took 2.8 s and cutting the traces 11 s, with at most 0.7 GB of memory in use. Loading the same file in one go took 15.6 s and 4.4 GB.

The one catch is a trace longer than the margin: it can get cut off at the end of a piece. If that happens, the summary under Detect / cut traces tells you how many traces were affected. If you hold the junction for minutes, make the margin longer.

5. Histograms tab

The 1D histogram of $\log(G/G_0)$ over all accepted traces is where the molecular conductance shows up as a peak. The 2D histogram (log G against distance) shows how far junctions get stretched at each conductance. Both are redrawn automatically whenever the traces change.

5.1 Settings

Setting	Meaning	Default
traces	All accepted traces, or one cluster from the Clustering tab	all
log G min / max	Conductance range of both histograms	-6.5 / 1
bins / decade	Conductance resolution	100
distance min / max, distance bins	Range and resolution of the 2D histogram	-0.5 / 2 nm, 200
normalization	`per trace` divides by the number of traces, so different data sets can be compared. `counts` is raw counts, `density` has area 1	per trace
2D colormap, colour max	Colour scale of the 2D histogram. The top of the scale is a percentile of the filled bins, so a few very full bins don't wash everything out	viridis, 99.5
overlay clusters in 1D	If there are clusters and *all* is selected, each cluster's histogram is drawn too, scaled by its share of the traces	on

5.2 Fitting the peak

Set *fit from* and *fit to* around the peak (keep away from the G_0 peak and the noise floor), choose the number of Gaussians and whether you want a constant baseline, and click Fit peaks. The fit is a sum of Gaussians in log G plus the optional constant:

$$N(x) = c + \sum_k A_k \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right), \qquad x = \log_{10}(G/G_0)$$

The starting values are the highest maxima in the range. For every peak you get the centre with its error, the conductance $G = 10^{\mu}$ in G_0, the width σ and the FWHM ($2.355\ \sigma$) in decades. The fit is drawn in red with the conductance written next to each peak.

For every peak the text also gives the dwell time, which is how long a trace spends in that conductance state on average. It comes from the area under the fitted peak (Rashid et al. 2025, SI 2.7):

$$\tau = \frac{A}{w} \cdot \frac{1000}{f_s}\ \mathrm{ms}, \qquad A = a\,\sigma\sqrt{2\pi}$$

Here A is the area of the Gaussian (amplitude a, width σ) in the per-trace histogram, w is the bin

width and f_s the sampling rate, so A/w is simply the number of samples per trace that fall inside the peak. This only works with *normalization* set to `per trace` or `counts`. This was checked on simulated data with 70 % molecular traces, 0.4 nm plateaus and 2 nm/s pulling: the expected dwell time is 0.7 x 0.2 s = 140 ms, and the fit gives 135-146 ms.

The program remembers the last fit; the Flicker noise tab uses it for the peak window (9.3).

If a peak has a shoulder or a long tail, two Gaussians usually describe it better than forcing one. And when you compare samples, compare the per-trace histograms, not raw counts.

6. Plateau length, junction statistics and change points

This chapter covers three tabs: Plateau length (6.1-6.3), Junction statistics (6.4) and Change points (6.5).

The plateau length is how far a junction can be pulled while the molecule is still bridging the gap. Compared with the length of the molecule it tells you whether junctions break when the molecule is fully stretched. The tab measures it in up to three conductance windows and also fits the tunnelling decay, which I use to calibrate the distance.

6.1 Settings

Setting	Meaning	Default
traces	All accepted traces or one cluster	all
window 1 from / to	Conductance window (log G), normally the peak +/- 1-2 σ	-4.5 / -2.5
window 2, window 3	More windows, e.g. for a second molecular state; empty = off	off
length method	`span` or `count`, see below	span
histogram max, bins	Range and resolution of the length histogram	2 nm, 60
ignore lengths below	Leaves traces without a plateau out of the statistics	0 nm
tunnelling fit from / to	log G range for the tunnelling fit (6.3)	-5 / -1
expected β (1/nm)	Decay constant you expect, for the calibration in 6.3	20

With `span`, the length runs from the first point below the upper limit to the last point above the lower limit, so short jumps out of the window don't make the plateau shorter. That's the usual stretching length. `count` multiplies the number of points inside the window by the distance step, so jumps out of the window are not counted.

6.2 Results

There's one length histogram per window, and the table gives the number of traces, mean, median, standard deviation, skewness, kurtosis, geometric mean, min and max for each.

Keep in mind that the real junction is longer than the measured length by the snap-back after the gold contact breaks, roughly 0.5 nm for gold. Add that before comparing with the molecule.

6.3 Tunnelling decay and distance calibration

For every trace log G is fitted linearly against distance in the tunnelling range, which gives β in G ~ exp(-β z). The right plot is the β histogram, the median is written under the table.

For gold in vacuum β is about 20 per nm (one decade of conductance per ~0.1 nm), in solvents it's

lower; put your value in *expected beta*. The label then tells you what to multiply the *displacement ratio* (4.5) by, measured β / expected β. Change the ratio and detect the traces again.

6.4 Junction statistics tab

The 1D histogram tells you where the molecular peak is, but not how many traces it came from. A peak made by 90 % of the traces and one made by 10 % can look almost the same once you normalise per trace. So this tab goes through the traces one by one and asks three questions: did a molecule bridge the gap, for how long, and at what conductance? The answers give you the junction yield, the fraction of traces with a molecular plateau (Kamenetska et al. 2009), and they also show whether the yield or the conductance drifted while you were measuring.

A trace counts as a junction if it stays flat inside the molecular window for at least *min plateau*, where flat means the local slope |d log G / dz| is below *max slope*. The slope test is there because otherwise every trace counts. A pure tunnelling trace falls through a two-decade window in about 0.1-0.2 nm, and with the plain `span` length from 6.1 that already looks like a short plateau. Tunnelling drops by β / ln 10 decades per nm, which is about 9 decades/nm for gold in vacuum and 4-5 in solvents, while molecular plateaus usually fall by well under one decade per nm. The default of 2 decades/nm sits in between.

The slope at a point is the change in log G between z - w/2 and z + w/2, divided by w, with w set by *slope over*. Points too close to the ends of a trace for that interval to fit never count as flat.

For every trace this gives a flat length (the summed distance of all flat points inside the window), a plateau conductance (the median log G of those points), and a yes or no: does the flat length reach *min plateau*?

Setting	Meaning	Default
traces	All accepted traces or one cluster	all
molecular window from / to	log G range where you expect the molecule	-4.5 / -2.5
flat if slope below	Slope limit in decades/nm	2
slope over	Distance over which the slope is measured	0.05 nm
junction if flat length above	Minimum flat length for a junction	0.1 nm
traces per block	Consecutive traces grouped for the plots against trace number	100
conductance bins	Bins of the plateau conductance histogram	60

Use peak from histogram fit sets the window to the fitted peak centre +/- 3 σ, so fit the peak in the Histograms tab first. Then click Calculate.

The text under the button gives the number of traces with a plateau, the yield with its binomial standard error sqrt(p(1-p)/N), and the median plateau conductance and length. If there are at least three blocks, it also gives the conductance drift in decades per 1000 traces (a straight line through the block medians) and how far the block yields spread. The plots are:

Plot	What it shows
Plateau conductance histogram	One entry per junction: the median log G of its plateau. Narrower than the 1D histogram, because the tunnelling background is gone
Conductance vs trace number	Every plateau conductance plus the median of each block. A slope means the junction changed during the run (electrode wear, solvent evaporating, the surface filling up)
Yield vs trace number	Yield per block with error bars and the overall yield as a dashed line
Flat plateau length histogram	Flat length of every trace, with the threshold marked. A clear gap near the threshold means the setting is sensible

Plot	What it shows
Length vs conductance	One point per junction. A tilt shows that longer-stretched junctions sit at another conductance, e.g. two binding geometries

To choose the settings, start with the flat length histogram. Tunnelling-only traces pile up near zero and the molecular traces form a second hump, so put *min plateau* in the dip between them. If there's no dip, the yield depends on where you put the threshold, and you should say which one you used. The method was checked on simulated data (12): with 70 % molecular traces it gives 68.5 +/- 3.3 %, and with 30 % it gives 29.0 +/- 3.2 %.

Keep in mind that the yield is per accepted trace. Traces rejected in Data & Traces (4.4) aren't counted at all, so looser or stricter cutting limits change the yield. Always give it together with the cutting limits.

6.5 Change points tab

The histogram and the junction statistics look at where a trace spends its time. Change-point detection asks a different question: where exactly does the conductance of a trace change? The tab has two methods. The first finds the moment the junction ruptures and takes the plateau conductance from everything before it. The second splits each trace into flat levels, so traces with two or more steps become visible.

The rupture method follows Hamill et al. (2024, ref. 38), who borrowed the Chow test from econometrics (ref. 39). Inside the window from *window: log G below* down to *window: log G above* (the first point below the upper limit to the first point below the lower one), log G against distance is described once by one straight line and once by two lines that meet at a break point k. The Chow statistic compares the two descriptions:

$$F(k) = \frac{(S_1 - S_2(k))/p}{S_2(k)/(n-2p)}, \qquad p = 2$$

S1 is the residual sum of squares of one line, S2(k) that of the two lines, and n the number of points. The break with the largest F is the rupture. The plateau conductance of the trace is the median log G before the break, and the length to rupture is the distance from the start of the window to the break. A trace counts as a junction if F reaches *Chow F threshold* and the line before the break is flatter than *max plateau slope*. The slope test matters as much as F: a pure tunnelling trace also gives a large F, because its slope changes a little along the way, but both of its lines are steep (4-9 decades/nm), while a molecular plateau falls by well under one decade per nm. The F threshold is a value of F, not a p-value, because taking the best of many break points makes the textbook F distribution far too optimistic.

The point of this method is that each junction contributes exactly one conductance, measured over its own plateau, and the tunnelling background below the plateau doesn't enter. Hamill et al. found that the spread of these values is about half the width of the peak in the 1D histogram, so the molecular conductance comes out sharper.

The step method uses PELT (Pruned Exact Linear Time, ref. 40) to describe log G as a sequence of flat levels. It finds the segmentation that minimises

$$\sum_{\text{segments}} \frac{\text{SSE}}{\sigma^2} + \lambda\, 2 \ln n \times (\text{number of change points})$$

where SSE is the squared deviation of the points from their level, σ the noise of the trace and lambda the *PELT penalty factor*. σ comes from the median absolute difference of neighbouring points, but

is never taken smaller than *noise at least*. The result is exact (the same as trying every possible segmentation), and the pruning keeps it fast. To keep it fast also for long traces, each trace is first averaged in blocks down to at most *max points per trace* points, which moves a step by at most one block. Neighbouring levels closer than *merge levels closer than* are joined. A tunnelling ramp gets cut into a staircase of short levels by such a fit, so only levels that are flat themselves (slope below *max plateau slope*) and at least *min level length* long count as plateau levels. The idea of looking at a variable number of change points per trace comes from Vaidya, Dyer and Monti (ref. 41).

Setting	Meaning	Default
window: log G below / above	Part of the trace that is analysed	-2 / -5.5
min segment (points)	Shortest piece on either side of a break	10
Chow F threshold	F needed for a rupture	10
max plateau slope (dec/nm)	The part before the rupture, and every plateau level, must be flatter	2
PELT penalty factor	lambda; larger gives fewer steps	1
PELT: max points per trace	Block averaging before PELT	400
PELT: noise at least (decades)	Lower limit of σ	0.05
PELT: merge levels closer than (dec)	Neighbouring levels closer than this are joined	0.15
PELT: min level length (nm)	Shorter levels don't count as plateaus	0.05
bins / decade	Histogram resolution	40

Click Find change points. The text gives the fraction of traces that rupture from a plateau, the median and mean plateau conductance with its standard error and spread, the median length to rupture, and how many plateau levels the traces have (0, 1, 2, ...). The plots are:

Plot	What it shows
Example trace	The trace chosen under *show trace* with the rupture (dashed), the plateau conductance (dotted) and the PELT levels (plateaus coloured, ramp pieces grey)
Rupture conductance	Histogram of the plateau conductance, one value per junction
Chow F	
Length to rupture	Histogram of the distance from the window start to the rupture
PELT plateau levels	Histogram of all plateau levels, weighted by their length
Step heights	Height of the steps between neighbouring plateau levels of a trace; a peak here means a well-defined second state

The Chow test assumes one rupture. For a trace with two plateaus it finds the first big drop, and the plateau conductance is the median of everything before it; the PELT levels show both. To choose the thresholds, look at the Chow F plot: junctions sit at small slope and large F, blank traces at large slope. If the two groups are well apart, the exact thresholds don't matter.

On simulated data the method finds 70.0 % junctions when 70 % of the traces have a plateau and 31.5 % when 30 % do, with a median plateau log G of -3.55 (true value -3.5, spread 0.25). PELT gives the result of an exhaustive search on every test series, and on a synthetic trace with two plateaus at log G = -2.9 and -3.9 (0.3 nm each) it finds both levels at the right height and length.

The files are `<name>_ruptures.txt` (per trace: break position, F, plateau log G, length, slopes, junction yes/no), `<name>_step_levels.txt` (every PELT level with its length, slope and whether it counts as a plateau), `<name>_step_level_histogram.txt` and `<name>_changepoints_summary.txt`.

7. Correlation tab

The correlation map shows which conductance values tend to turn up in the same trace. That way you can see things the 1D histogram averages away, for example a molecule going through two binding geometries one after the other.

For each trace r the program makes its own histogram N_r(G) on a common set of bins, and then calculates the Pearson correlation between every pair of bins over all traces (Makk et al. 2012):

$$C(G_i, G_j) = \frac{\langle \delta N_r(G_i)\, \delta N_r(G_j) \rangle_r}{\sqrt{\langle \delta N_r(G_i)^2 \rangle_r \langle \delta N_r(G_j)^2 \rangle_r}}, \qquad \delta N_r = N_r - \langle N_r \rangle_r$$

Setting	Meaning	Default
traces	All accepted traces or one cluster	all
log G min / max	Range of the map	-6 / 0.5
bins / decade	Coarser than in the histograms, so each trace puts enough counts in each bin	20
colour limit	Colour scale goes from minus to plus this	0.3
colormap	Diverging scale, red positive and blue negative	RdBu_r

Click Calculate. The diagonal is always 1, and the second plot is the mean per-trace histogram so you know where the peaks are.

How to read it: a positive region off the diagonal at (G1, G2) means traces that stay long at G1 also stay long at G2, like two plateaus that come together. Negative means the two exclude each other, e.g. two different binding geometries. Around zero there's no relation, and empty bins have no variance and stay blank.

You need a few hundred traces. With fewer the map is mostly statistical noise (of the order of 1/sqrt(number of traces)).

8. Clustering tab

Clustering sorts the traces into groups of similar shape without needing a reference trace, for example molecular traces vs. pure tunnelling, or two binding geometries. Once the traces are clustered, the other tabs can work on one cluster at a time.

8.1 What it does

Each accepted trace is first turned into a vector of fixed length. By default that's a small 2D histogram (distance x log G), flattened and normalised to sum 1, so traces of different length and sampling can be compared (same idea as Lemmer et al. 2016 and Cabosart et al. 2019). These vectors are centred and reduced with PCA, the clustering runs on the first components, and finally the clusters are numbered either by median conductance (highest first) or by size.

8.2 Settings

Setting	Meaning	Default
feature	`2D histogram` (what I'd use), `1D histogram` (log G only) or `resampled trace` (log G on a fixed distance grid)	2D histogram
log G min / max, log G bins	Conductance grid of the feature	-6 / 0.5, 28

Setting	Meaning	Default
distance min / max, distance bins	Distance grid of the feature	-0.2 / 1.5 nm, 28
resample points	Grid size for `resampled trace`	128
PCA components	How many components the clustering uses; 0 = raw features	10
algorithm	see 8.3	k-means
number of clusters	k	3
random seed	Keeps k-means, Gaussian mixture and spectral results the same from run to run	0
order clusters by	`conductance` or `size`	conductance
k range for scan	Which k the silhouette scan tries	2, 8

8.3 Which algorithm

Algorithm	Good for	Note
k-means	A first look, compact groups of similar size	Fast, assumes round clusters
gaussian mixture	Groups with different spread or orientation	Elliptical clusters, slower
agglomerative	Hierarchical structure	Ward linkage
spectral	Long or curved groups	Gets slow above ~5000 traces

8.4 Running it

Run clustering uses the k you set; Run + scan k also computes the silhouette score for each k in the scan range. The plots are the scree plot, the PC1-PC2 scatter coloured by cluster, the 1D histogram of each cluster, the silhouette scan (only after Run + scan k) and a 2D histogram for every cluster. The text lists the size of each cluster and the silhouette score (-1 to 1, higher means better separated). Clear clusters removes the labels again.

Don't just take the k with the highest silhouette score. With noisy data it often keeps going up with k. I take the smallest k where the cluster 2D histograms clearly look different and make physical sense; typically that's one tunnelling cluster and one or more molecular ones.

To use a cluster, pick `cluster N` in the *traces* box of Histograms, Plateau length or Correlation. When you save results, the histograms of every cluster are written automatically. Detecting the traces again removes the clusters.

9. Flicker noise, events and blinking

This chapter is about conductance against time: the flicker noise (9.1-9.7), the Events tab for flickering and mechanical switching (9.8), and the Blinking tab for fixed-gap I(t) measurements (9.9).

The flicker noise is used to tell through-bond transport from through-space tunnelling. The noise power NP goes with the mean conductance as NP $\sim G^n$. If n is close to 1, the coupling is through-bond and the current goes through the molecular orbitals. If n is close to 2, it's through-space: electrons tunnel directly between the electrodes, and since G $\sim$ exp(-β d), fluctuations of the gap give dG $\sim$ G and so NP $\sim G^2$. Anything in between is a mix.

Two methods are implemented, the bell-shaped 2D histogram from Adak et al. (2015) and the Theil-Sen method with a stationarity test from Morris et al. (2025). They run on the same data and you always get both results.

9.1 What data you need

The noise is measured on G(t) traces taken while the junction is held still. Typically you pull until you're on the molecular plateau, stop the piezo for 0.1-2 s, then carry on. The Data & Traces tab finds these holds by itself (4.3), so detect the traces first.

If your data has no holds, set *segments* to `plateau windows`; then windows inside the breaking traces are used. These aren't as stationary as real holds, so be careful with the result.

Morris et al. found that n stops depending on the sampling rate above about 5 kHz for their 90 ms traces. Shorter holds probably need faster sampling.

9.2 Noise power

Each analysis window is either a whole hold (after trimming) or a fixed-length piece of one. The first and last few milliseconds are cut off (*trim start*, *trim end*) because the junction is still relaxing right after the piezo stops. Then the PSD of G (in G_0) is calculated, with Welch's method or as a periodogram. The mean of the trace is subtracted first (*detrend* `constant`); if you don't, a trace with a step in it gives a fake $1/f^2$ spectrum. Finally the PSD is integrated over the band from *integrate from* to *integrate to*:

$$NP = \int_{f_1}^{f_2} S_G(f)\, df \qquad [G_0^2], \qquad f_1 = 100 \text{ Hz},\ f_2 = 1000 \text{ Hz}$$

100-1000 Hz is the usual choice: below that there's mechanical noise, above it amplifier noise. Whatever you choose, keep it the same for all molecules you want to compare.

9.3 Which windows are kept

The checks below are done in this order, and every rejected window is counted under its reason in the result text.

1. Conductance range: the mean log G of the window has to be between *accept log G from* and *to*. This removes the noise floor and gold contacts.
2. Peak window: if *peak centre* and *peak sigma* are set, the mean conductance of the first and the last *end fraction* (5 %) of the window (or the first and last *end points* points, if that is above 0) both have to be within centre +/- *n sigma* x σ (default 2). That way only junctions that stay on the molecular plateau are used. Peak from histogram fit copies centre and σ from the last fit in the Histograms tab. If you publish, say which n σ you used.
3. Kurtosis (optional): throws out windows with too high excess kurtosis, i.e. with switching or spikes.
4. Drift (optional): throws out windows where the averaged log G at the start and at the end differ by more than the limit.
5. Stationarity (optional, on in the Morris preset): Augmented Dickey-Fuller test on log10|G| with a constant term, maximum lag ceil(12 (N/100)^(1/4)) and the lag chosen by AIC. The window is kept if p <= *ADF significance* (0.05), i.e. if the unit-root hypothesis is rejected. The implementation was checked against statsmodels and gives the same statistic, lag and p-value.

With *subtract instrument floor* on, the mean PSD of the windows below *floor = windows below log G* is integrated over the same band and subtracted from every NP.

9.4 How n is determined

All three estimators are always calculated. *estimator for n* only decides which one is used for the bell plot and the interpretation text.

Estimator	How	Comment
Theil-Sen (default)	Median of the slopes between all pairs of points in log NP vs log G, which is the same as the n where Kendall's τ of (log G, log NP/G^n) is zero. Error from Sen's 95 % confidence interval	Up to 29 % outliers don't change it. This is what Morris et al. recommend
OLS	Least-squares slope of log NP vs log G, the same as the n where the Pearson correlation is zero	Sensitive to outliers and long tails, tends to give n too high
2D Gaussian	n where the fitted 2D Gaussian (the bell) of log(NP/G^n) vs log G has zero correlation	The graphical method of Adak et al. 2015

The OLS error is also checked with a bootstrap (*bootstrap samples*). If there are more than 6000 windows, Theil-Sen uses a random 6000 of them, otherwise it gets slow.

9.5 Presets

Preset	What it sets
Morris et al. 2025 (ADF + Theil-Sen)	5 ms trimmed at both ends, one window per hold, mean-subtracted one-sided periodogram (rectangular window), 100-1000 Hz, ADF with α 0.05, ends within +/- 2 σ of the peak, no drift or kurtosis cut, Theil-Sen
Adak et al. 2015 (2D Gaussian bell)	100 ms windows, Welch PSD, 100-1000 Hz, 2D Gaussian estimator
Rashid et al. 2025 (Pearson r scan)	First 10 ms of each hold dropped, the rest is one window; first and last 100 points within +/- 1 σ of the peak; DFT squared (periodogram) integrated 100-1000 Hz; no ADF, drift or kurtosis cut; n where the Pearson r of log(NP/G^n) and log G is smallest, scanned 0.3-2.3 in steps of 0.01 (the OLS estimator)

The presets don't touch your conductance range. After the Morris preset, press Peak from histogram fit so the peak window is set.

9.6 Plots

Where	What
Top left	The segment chosen in *show segment*, accepted windows green, rejected red
Top middle	Mean PSD of the accepted windows, the floor PSD, the integration band and a $1/f^\alpha$ fit
Top right	log NP vs log G with the OLS and Theil-Sen lines
Bottom left	The bell: 2D histogram of log(NP/G^n) vs log G at the chosen n, with the fitted Gaussian as contours
Bottom middle	Pearson r, Kendall τ and the 2D Gaussian correlation against n; each estimator is where its curve crosses zero
Bottom right	Residuals of the OLS and Theil-Sen fits; long tails mean outliers

9.7 What to report

n with its error, which estimator, how many traces were accepted, the frequency band, the trimming, the n σ window and whether the ADF test was on. All of this is in `*_Noise_summary.txt` (13). If OLS and Theil-Sen differ by more than their errors there are outliers in the data; I'd trust Theil-Sen and have a look at the residual plot.

The analysis was checked on simulated data with n = 1.5, which gives n_TSE = 1.52 +/- 0.02. After adding 5 % outlier windows, Theil-Sen gave 1.40 and OLS 1.33.

The Rashid et al. 2025 preset was checked on five simulated data sets with 160 ms holds on the plateau and n = 1.5. It accepted 55-73 holds per set and gave n = 1.53 on average; the Morris preset gave 1.59 on the same data. The text also reports the *minimum |Pearson r| on the scan grid*, which is how we read n off the scan in the paper. It agrees with the OLS n to within the grid step.

9.8 Events tab: flickering and mechanical switching

When a junction switches between two states during a trace, you see it as steps in G(t). These are either short back-and-forth steps (flickering) or one large jump, for example when stretching turns one isomer into the other (a mechanical event). The Events tab finds these steps the same way we did in Rashid et al. (2025, SI 2.6).

First G(t), in G_0, is smoothed with a Butterworth low-pass (200 Hz, order 2) and then a Savitzky-Golay filter (order 20 with 20 side points, so 41 points wide). Without this the noise alone would produce steps. The smoothed trace is then differentiated with the central difference dG/dt = (G[i+1] - G[i-1]) / (2 dt), and every peak of the derivative above a threshold is an event: a mechanical event above the *mechanical threshold* (0.002), and a flicker between the *flicker threshold* (0.0007) and that. Events that happen while the junction is outside the molecular range (*valid from / to log G*), so in metal contact or down at the noise floor, are false positives and get dropped.

Each trace ends up as *mechanical* (at least one mechanical event), *flickering* (only flickers) or *quiet*.

Setting	Meaning	Default
segments	`traces` (accepted breaking traces, against time) or `holds`	traces
low-pass, order	Butterworth filter (applied forwards and backwards, so no time shift)	200 Hz, 2
Savitzky-Golay side points, order	Smoothing after the low-pass	20, 20
derivative dt (s)	dt in the derivative; the thresholds are on this scale	0.02
peak width, peak detection	See below	50, quadratic fit
count	`rises`, `drops` or `both`	rises
flicker / mechanical threshold	Thresholds on dG/dt	0.0007 / 0.002
valid from / to log G	Molecular range	-6.5 / -1
level before/after (ms)	The conductance before and after an event is the median over this time	20
hold histogram: time bins, bins / decade	For the conductance-time histogram of the holds	200, 30

Three details of the original LabVIEW code aren't in the paper, so I went with what LabVIEW does by default. If your own settings were different, change these:

- dt: LabVIEW's derivative takes dt as an input. With dt = 20 ms and 10 kHz data the peak of dG/dt is about 2.2 times the step height (measured on synthetic steps), so the thresholds of 0.0007 and 0.002 correspond to steps of roughly 3 x 10^{-4} and 9 x 10^{-4} G_0.
- Peak width: LabVIEW's Peak Detector fits a quadratic over *width* points (50) and compares the fitted maximum with the threshold. That's *peak detection* = `quadratic fit`. The other option, `minimum width`, only keeps peaks at least *width* samples wide at half height. Be careful with it: after a 200 Hz low-pass the derivative peaks are only 20-30 samples wide at 10 kHz, so `minimum width` with 50 throws out almost everything.
- Polarity: the Peak Detector only finds maxima, hence *count* = `rises`, and a flicker up and back down counts once. Choose `both` if you want every step counted.

One technical note: scipy's own Savitzky-Golay coefficients fall apart at order 20 (they come out around 10^{-17}), so the program builds the filter from a QR decomposition in the Legendre basis instead. That matches scipy to 10^{-14} at low order and stays exact at order 20.

The plots show an example trace with the filtered signal and the events marked, its derivative with both thresholds, the share of quiet, flickering and mechanical traces, the number of flickers per trace, and a transition map of log G before against log G after each event. With `holds` you also get the 2D conductance-time histogram of all holds, with the most probable log G at each time (a Gaussian fit per time bin) and its σ as error bars. That's the plot we used for the mechanical modulation in the paper (Fig. S23): hold the junction, move the piezo back and forth with a square wave, and watch the most probable conductance follow.

On synthetic traces it picks up a 4 x 10^{-4} G_0 flicker at G ~ 10^{-3} G_0 as a flicker and a jump from 10^{-6} to 1.6 x 10^{-3} G_0 as a mechanical event, and finds nothing in a flat trace with 3 % noise.

9.9 Blinking tab: fixed-gap I(t) measurements

In a break junction the electrodes keep moving. In a blinking experiment they don't: the STM tip (or the MCBJ electrodes) is held at a fixed distance, a little too far apart for anything but tunnelling, and the current is recorded over time. Every now and then a molecule bridges the gap, and the current jumps up to the molecular level. It drops back when the molecule leaves or the contact breaks. The current switches between two levels like a telegraph signal, which is why this is called blinking, or the I(t) method (Haiss et al., refs. 42 and 43). It gives the single-molecule conductance without stretching the junction, and the times give two numbers a pulling experiment can't: how long a molecule stays in the junction and how often one arrives.

The tab works on a whole recording (*segments* = `recording`, the file chosen above; for a big TDMS file only the part between *from* and *to*), or on the holds found in Data & Traces (`holds`). For each segment:

1. log G is smoothed with a moving average over *smooth*.
2. The baseline, the tunnelling current without a molecule, is a low percentile (*baseline percentile*) of log G in blocks of *baseline window*, interpolated in between. A percentile instead of a mean keeps the blinks from pulling the baseline up, and the blocks let it follow a slow drift of the gap. The percentile has to be lower than the fraction of time without a molecule; 20 % works as long as a molecule is in the gap less than 80 % of the time.
3. A blink starts when log G rises more than *up threshold* above the baseline and ends when it falls below *down threshold*. Using two thresholds (hysteresis) means noise on top of a molecular level doesn't split it into several blinks. Gaps shorter than *min off* are closed, and blinks shorter than *min on* are dropped. Blinks cut off by the start or end of a segment are left out too, because their length isn't known.
4. For every blink the program keeps its start, its length (the on time), its conductance (median log G while on), the baseline under it and the jump height. Blinks whose conductance is outside *blink log G from / to* (metal contacts, noise) are dropped.

From all the blinks the tab takes the blink conductance (a Gaussian fit to the histogram of the blink levels, with the error of its centre) and the on-time and off-time lifetimes. For exponentially distributed times above a cut-off t_min the maximum-likelihood lifetime is

$$\tau = \frac{N}{N-1}\,(\bar{t} - t_{\min})\,, \qquad \delta\tau = \tau/\sqrt{N}$$

with t_min the shortest time measured, so blinks too short to be detected don't bias τ. tau_on is the lifetime of the molecular junction, and 1/tau_off is the rate at which molecules arrive, which depends on the concentration and the gap. The text also gives the blink rate (blinks per second) and

the fraction of the time a molecule is in the gap.

Setting	Meaning	Default
segments	`recording` or `holds`	recording
smooth (ms)	Moving average before the detection	1
baseline window (s), baseline percentile	Block length and percentile of the baseline	0.5, 20
up / down threshold (decades)	Blink starts above up, ends below down (relative to the baseline)	0.6 / 0.3
min on, min off (ms)	Shortest blink; shortest gap between two blinks	1, 0.5
blink log G from / to	Window for the blink conductance	-6 / -1
bins / decade, time bins	Histogram resolution	40, 40
aligned: before / after (ms)	Time range of the aligned histogram	10 / 50

The plots are an example segment with the smoothed signal, the baseline, the up threshold and the blinks shaded, the blink conductance histogram with its Gaussian, the on-time and off-time histograms on a log scale with the exponential fits, the blink conductance against the on time, and a 2D histogram of log G against the time from the blink start. The last one shows how sharp the jump is and whether the level relaxes during a blink. Add simulated makes a 20 s recording with known answers.

Two things to watch. First, the up threshold has to sit clearly between the noise of the baseline and the jump height. The median jump height in the text tells you where the jumps are; if many blinks have heights close to the threshold, the threshold is cutting into the noise. Second, the on-time histogram should look like a straight line on the log scale. A bend at short times usually means *min on* or the smoothing is too short; two slopes mean two kinds of junction.

On the simulated recording (blinks at log G = -3.5 with σ 0.15, tau_on = 20 ms, tau_off = 100 ms, 0.08 decades of noise and a drifting baseline) the tab finds 171 of the 172 blinks, a blink conductance of -3.53 +/- 0.01, tau_on = 20.2 +/- 1.5 ms and tau_off = 95 +/- 7 ms.

The files are `<name>_blinks.txt` (every blink: segment, start, duration, log G, baseline, height), `<name>_blink_off_times.txt`, `<name>_blink_histogram.txt` and `<name>_blinking_summary.txt`.

10. I-V, EGaIn and Thermopower tabs

These three tabs all deal with current against bias or temperature rather than conductance traces. The I-V tab (10.1-10.4) cuts a recording with bias sweeps into forward and backward I-V curves, makes histograms of them and, if you want, fits a transport model. The EGaIn tab (10.5) does the statistics of large-area junctions, and the Thermopower tab (10.6) gets the Seebeck coefficient from thermovoltages.

10.1 Finding the sweeps

For this the bias has to be recorded over time, so you need a TDMS file with a bias channel, or a text or Igor file with a *voltage column* (4.2). With a constant bias you just get "No bias sweeps were found".

The bias is smoothed and its turning points found (peaks and valleys at least *min sweep amplitude* apart). Every monotonic piece between two turning points is one curve, forward if the bias goes up and backward if it goes down. The current of each curve is smoothed with a Savitzky-Golay filter and dI/dV is the ratio of the filtered derivatives of I and V.

Setting	Meaning	Default
file	Which recording	first
from / to (s, big files)	For a TDMS file read in pieces (4.7): the part of the file to analyse; empty = all	empty
min sweep amplitude (V)	Smallest bias swing that counts as a sweep	0.05
smoothing points, polynomial order	Savitzky-Golay window and order for the current and dI/dV	11, 2
curve must reach V below / above	Only keeps curves that cover this range; empty = off	off
V bins, I bins	Resolution of the 2D histograms	100, 100
log |I| histogram	Histograms log10 |I| instead of I	on

The top left plot shows up to 200 curves (forward blue, backward orange) together with the averaged forward and backward curves. The other three are 2D histograms against V of log I, of log(G/G_0) with G = I/V, and of log(dI/dV / G_0). The curves are saved as V/I column pairs in `IV_DATA/IV_Curves_F.txt` and `IV_Curves_B.txt` (13).

Only sweeps that cover the typical bias range go into the mean curves. The half sweeps at the start and end of a recording would shrink the common range to almost nothing, so they're left out of the average, but they still show up in the histograms and in the fits of single curves.

10.2 Single-level model

Most I-V curves of single-molecule junctions can be described by a single molecular level, sitting at energy ε_0 from the Fermi level of the electrodes and broadened by its coupling Γ to each electrode. The transmission is then a Lorentzian, and at zero temperature the Landauer current has a closed form:

$$I(V) = N\,G_0\,\Gamma\left[\arctan\frac{V/2-\varepsilon(V)}{\Gamma}+\arctan\frac{V/2+\varepsilon(V)}{\Gamma}\right], \qquad \varepsilon(V)=\varepsilon_0+aV$$

Energies are in eV and V in volts, so G_0 times an energy in eV is directly a current in A. Both electrodes are coupled equally ($\Gamma_L=\Gamma_R=\Gamma$), N is the number of molecules in parallel (keep it at 1 for single molecules), and a describes an asymmetric junction where the level moves with the bias. At low bias the model reduces to G/G_0 = N Γ^2 / ($\varepsilon_0^2+\Gamma^2$). Zotti et al. (2010), for example, used this form to compare anchoring groups.

Above 0 K the Fermi functions of the leads smear out the edges of the bias window, and you need the general Landauer form

$$I=\frac{2e}{h}\int dE\,T(E)\,[f_L(E)-f_R(E)], \qquad T(E)=\frac{\Gamma^2}{(E-\varepsilon)^2+\Gamma^2}$$

with f_L and f_R the Fermi functions at $\mu=+$V/2 and -V/2. This T(E) is the same as 2 π Γ_L Γ_R D(E) / Γ_{tot} with a Lorentzian density of states D(E) of width $\Gamma_{\mathrm{tot}}=\Gamma_L+\Gamma_R$, which is how we wrote it in Rashid et al. (2025). Γ here is $\Gamma_{L,R}$, the number given in Table S1 of that paper.

The program can do this integral in two ways. With `analytic` it uses the fact that a Lorentzian integrated against a Fermi function has a closed form in terms of the digamma function psi, which is exact and fast:

$$\int T(E)\,f(E-\mu)\,dE=\pi\Gamma\left[\frac{1}{2}-\frac{1}{\pi}\,\mathrm{Im}\,\psi\!\left(\frac{1}{2}+\frac{\Gamma-i(\mu-\varepsilon)}{2\pi k_B T}\right)\right]$$

With `energy grid` the integral is done numerically, the way we did it in the paper (Rashid et al. 2025, SI 2.4): from -10 to 10 eV in steps of 0.02 meV. To keep that fast, T(E) is integrated once and the Fermi functions come in through a single FFT convolution per evaluation. The two methods agree to better than 10^{-9} (checked at 0 and 300 K with $\varepsilon_0 = 1.1$ eV and $\Gamma = 11.3$ meV). Single curves are always fitted with the closed form. The grid is used for the mean and most probable curves and starts from the closed-form result, so even a grid fit takes less than a second.

Besides the mean forward and backward curves, the program also builds the most probable I-V curve. It bins all curves by bias, fits a Gaussian to the current distribution in each bin and takes its centre, with σ as the error bar. This is the blue line in Fig. S22 of our paper, and a few odd curves bother it much less than they bother the mean. It gets fitted as well; *weight by 1/sigma* weights each bias bin by its spread.

The Rashid et al. 2025 preset sets 300 K, the energy grid (-10 to 10 eV, 0.02 meV), the Levenberg-Marquardt algorithm, $\Gamma_L = \Gamma_R$, no asymmetry and N = 1. The SI doesn't state the temperature, so 300 K is an assumption; change it if you measured at another temperature. On a model curve made with the G1 values from Table S1 ($\varepsilon_0 = 1.10$ eV, $\Gamma = 11.3$ meV, 300 K) the preset gets back 1.100 eV and 11.30 meV.

For large-area junctions, *fit N too* lets the number of molecules float (the EGaIn tab uses this). N and Γ are then strongly correlated: on a model curve with N = 10^4 and $\Gamma = 50$ meV the fit returns N = 8.6 x 10^3 and $\Gamma = 54$ meV, while ε_0 comes back within 0.1 %. So report ε_0, and take N and Γ with a grain of salt.

To run the fit, click Fit single-level model + TVS after Analyse I-V. It's a least-squares fit of I (scaled by its maximum), started from four values of ε_0, and the best of the four is kept. Every single curve is fitted, up to *fit at most N single curves*, and so are the mean forward and backward curves.

Setting	Meaning	Default
fit only |V| below	Restricts the fit to low bias; empty = whole curve	empty
fit asymmetry a	Also fits a; off means a = 0 (symmetric junction)	off
molecules in parallel N	N in the formula	1
TVS: ignore |V| below	See 10.3	0.05 V
fit at most N single curves	Fitting takes a few ms per curve	500
temperature (K)	0 = zero-temperature closed form	0
mean-curve integration	`analytic` or `energy grid` (see above)	analytic
energy grid step, from, to	Grid of the numerical integral	0.02 meV, -10, 10 eV
fit algorithm	`trust region` (bounded) or `Levenberg-Marquardt` (unbounded)	trust region
fit N too	Let N float (ensembles)	off
most probable: bias bins, current bins	Binning for the most probable curve	40, 100
weight by $1/\sigma$	Weights for the most probable curve	off

A few things to keep in mind when you read the numbers. ε_0 always comes out positive, because HOMO and LUMO transport give the same I-V; to tell them apart you need thermopower (10.6) or theory. When the bias stays well below ε_0, the curve is nearly linear with a small cubic term, so ε_0 and Γ become strongly correlated and their errors (from the fit covariance) get large. The ε_0 vs Γ scatter shows this as a long diagonal streak, which means the data can't separate the two. Finally, the model assumes a single level that moves with the bias by no more than a V, and at *temperature* 0 it also assumes ε_0 and Γ are much larger than kT (about 25 meV); if they aren't, set the temperature.

When these assumptions fail, the model can still fit the curve nicely, but ε_0 and Γ no longer mean what they seem to.

10.3 Transition voltage spectroscopy (TVS)

Plot ln($|I|/V^2$) against 1/V (a Fowler-Nordheim plot) and the I-V curves of molecular junctions show a minimum. Beebe et al. (2006) called the bias where it happens the transition voltage V_t; it's the same as the maximum of $V^2/|I|$. V_t needs no fitting and is less sensitive to noise than the curvature of I(V), which is why so many people use it.

For each curve and each polarity the program averages I in 60 bias bins above *TVS: ignore |V| below*, finds the maximum of $V^2/|I|$ and refines it with a parabola through the three highest points. If the maximum sits at the edge of the measured range, the bias simply didn't go far enough, and that V_t is left empty.

In the single-level model with $\Gamma \ll \varepsilon_0$, the two transition voltages give ε_0 and a in closed form (Baldea 2012):

$$\varepsilon_0 = \frac{2\,|V_t^+ V_t^-|}{\sqrt{(V_t^+)^2 + (V_t^-)^2 + \frac{10}{3}|V_t^+ V_t^-|}}, \qquad a = \frac{(V_t^+ + V_t^-)\,\varepsilon_0}{4\,|V_t^+ V_t^-|}$$

For a symmetric junction this reduces to $\varepsilon_0 =$ (sqrt(3)/2) V_t, about 0.87 V_t, and that's what is used if only one polarity has a V_t. Both formulas were checked against the exact model: with $\varepsilon_0 = 0.8$ eV, $\Gamma = 10$ meV and a = 0.1 the curves give $V_t^+ = 1.19$ V and $V_t^- =$ -0.75 V, and from those the formulas give back $\varepsilon_0 = 0.800$ eV and a = 0.100.

It's worth comparing the ε_0 from TVS with the one from the fit. If they disagree a lot, the curves probably aren't single-level-like: there may be several levels, a strong asymmetry, or heating at high bias.

10.4 Plots and what to report

The fit and TVS add five plots to the I-V tab:

Plot	What it shows
SLM fit	Mean forward and backward curves (dots), the most probable curve with its σ, and the fitted model (lines)
Fowler-Nordheim	ln($|I|/V^2$) vs 1/V of the mean curves, positive bias solid and negative dashed, V_t as dotted lines
Transition voltage histogram	V_t^+ and $|V_t^-|$ of all curves
SLM parameters	ε_0 against Γ (log scale) for every fitted curve
ε_0 histogram	ε_0 from the fits and from V_t

When you report the results, give ε_0 and Γ with their spread over the single curves (the medians are in the text), say whether a was fitted, and give the bias range of the fit. For TVS, give both polarities. The numbers per curve are in `<name>_IV_models.txt`, the most probable curve with its fit in `<name>_IV_most_probable.txt`, and the summary in `<name>_IV_models_summary.txt` (13).

10.5 EGaIn (ensemble) tab

Large-area junctions (EGaIn tips, CP-AFM, crossbars) measure many molecules at once, and you usually make many junctions per sample. Their statistics differ from break junctions in one important way: the unit is the junction, not the sweep. A sample with 20 junctions and 20 sweeps on each is 20

independent measurements, not 400. The tab follows the statistics of Reus et al. (2012), the same as GaussFit (Chiechi group) and the EGaIn module of XMe (Hong group).

There are two file layouts. With *file layout* = `one junction per file`, each file is one junction, with a column for the bias and one for the current (or J). With `column pairs V, I, V, I, ...` one file holds many junctions: columns 0 and 1 are the bias and current of the first junction, 2 and 3 of the second, and so on. The pairs can have different lengths (the shorter ones just end in empty cells), a header line is fine, and the junctions get the file name with _1, _2, ... appended. Add files ... or Add folder ... makes a *data set*, which is normally one molecule or one sample. If you want β, type a molecular length for each data set into the table; nm, angstrom or the number of carbons all work, as long as every set uses the same unit. Add simulated makes fake data sets where you know the right answer.

When you click Analyse all, the current is first turned into a current density J = I / A, with A = π d^2 / 4 from the contact diameter or the area you typed in. A junction whose |J| ever reaches the *short threshold* is a short: it counts against the yield (working junctions / all junctions) and is left out of everything else. The bias is cut into sweeps where it turns around, and each sweep is put on a common bias grid (*bias grid step*), without 0 V, where log|J| makes no sense.

At every bias, the log10|J| values of all sweeps are histogrammed (*bins / decade*) and fitted with a Gaussian, and its centre <log|J|> and width $\sigma_{\log}$ are what you report, as in Reus et al. The confidence interval uses the number of working junctions n_j as the degrees of freedom (formula below), which is where the junction-not-sweep rule comes in.

For the rectification, sweeps are grouped into cycles that cover both polarities, and log R = log|J(+V)| - log|J(-V)| is histogrammed and fitted at every |V|. Each cycle also gets its V_{trans} (10.3). If you like, the single-level model (10.2) is fitted to the Gaussian-mean J(V), with N free (the number of molecules per cm^2 that carry the current).

Fit β needs at least two analysed data sets with lengths. It fits log10 <|J|> at *report at bias* against the length d (formula below), weighting each point by the standard error of its mean, $\sigma_{\log}$ / sqrt(n_j - 1). With more than two sets the error is scaled up by the scatter of the points if that's larger. β comes out per unit of the length you typed; Simeone et al. (2013) report 0.92 per carbon for alkanethiolates.

The confidence interval and the length dependence are

$$\mathrm{CI} = t_{1-\alpha/2,\, n_j-1}\,\frac{\sigma_{\log}}{\sqrt{n_j-1}}, \qquad \log_{10}|J| = \log_{10} J_0 - \frac{\beta\, d}{\ln 10}$$

Setting	Meaning	Default
file layout	`one junction per file` or `column pairs V, I, V, I, ...`	one junction per file
bias column, current column	Column number (0 = first) or header name (one junction per file only)	0, 1
current unit	A, mA, uA, nA, or A/cm2 if the column is already J	A
contact diameter (um), contact area (cm2)	Geometric contact; the area overrides the diameter	25 um, empty
short if |J| reaches (A/cm2)	Short threshold	100
bias grid step (V)	Bias grid of the statistics	0.05
log|J| bins / decade	Histogram resolution	10
CI: α	Confidence level 1 - α	0.05

Setting	Meaning	Default
report at bias (V)	Bias of the log|J| and R histograms and of β	0.5
single-level fit, temperature	Fit of the mean J(V)	on, 300 K

The plots are for the data set chosen in *show*: every sweep with <log|J|> +/- $\sigma_{\log}$, the 2D histogram of log|J| against bias, the log|J| histogram at the report bias with its Gaussian, the log R histogram, the V_{trans} histograms, the single-level fit and, once you've clicked Fit β, log|J| against length.

The tab was checked on simulated sets of 30 junctions with $\sigma_{\log} = 0.4$, a rectification of log R = 0.3 and $\beta = 0.9$. It finds <log|J|> within its confidence interval of the true value, log R = 0.30 +/- 0.08, all the shorts, and $\beta = 0.904$ +/- 0.016. Repeating it 20 times with three sets of 20 junctions gave $\beta = 0.92$ on average, with a scatter of 0.087 and an average reported error of 0.095. So the error bar is realistic: 65 % of the runs land within 1 σ, where 68 % is expected.

In a paper, give <log|J|> and $\sigma_{\log}$ at a stated bias, the number of junctions and of sweeps, the yield, the contact area and how you measured it, and the confidence interval with α. Remember that the geometric contact area isn't the electrical one (Simeone et al. 2013 estimate the electrical area at 10^{-4} of it), so J_0 can only be compared between data sets measured the same way.

10.6 Thermopower tab

Heat one electrode of a molecular junction and a thermoelectric voltage appears across it. Its sign and size give the Seebeck coefficient S, which tells you whether the HOMO or the LUMO is closer to the Fermi level: S > 0 means hole transport through the HOMO, S < 0 electron transport through the LUMO. This is exactly what the single-level fit can't tell you (10.2). The tab does the standard analysis of Reddy et al. (2007) and Widawsky et al. (2012): you give it the thermovoltage and the temperature, and it makes the histograms and the fit.

The input is text or Igor files with the thermoelectric voltage in one column and the temperature in another, one row per sample. Usually a file is one trace, so one junction, but since the temperature is read for every sample, a file can just as well hold traces at several temperatures. The temperature column can hold the temperature difference dT itself, or the temperature of the heated electrode, in which case dT = T - *T reference* (the cold side, usually room temperature). If you don't have a temperature column at all, each data set gets one dT that you type in when you load it. An optional conductance column (G/G_0 or log G/G_0) lets you keep only the samples on the molecular plateau (*keep log G from / to*), and then S is also plotted against log G.

The voltage is converted to microvolts and divided by the *amplifier gain*; *invert voltage sign* flips it if your wiring measures the opposite polarity. The samples are then grouped by dT in steps of *dT step*. With *histogram of* = `file means` each file gives one mean voltage per dT, so one value per junction as in Reddy et al.; with `samples` every sample counts. At every dT the thermovoltage is histogrammed and fitted with a Gaussian, whose centre is the most probable thermovoltage V(dT). A straight line through V(dT), weighted by the error of each centre, gives the Seebeck coefficient from its slope:

$$S = -\frac{dV}{d(\Delta T)} + S_{\mathrm{electrode}}$$

On top of that, every sample (or file) with dT not 0 gets its own S_i = -V_i/dT_i + S_electrode, and these are histogrammed and fitted too, and plotted against log G if there's a conductance column.

The error of each peak comes from a bootstrap over the files (*bootstrap* resamplings). That's needed because the samples of one trace aren't independent, and a Gaussian fitted to a histogram scatters more than the plain standard error would suggest. On simulated data (30 repeats, 20 junctions per

dT, S = -6 uV/K) the mean S was -6.05 and 80 % of the results fell within their error, so the error is honest, if anything a little conservative.

S of the electrodes is added to -dV/d(dT). With the default of 0 you get S relative to the electrode leads. Reddy et al. add the Seebeck coefficient of gold; if you want the absolute value for the junction, enter the one for your electrode metal and temperature.

Setting	Meaning	Default
voltage column, voltage unit	Column of the thermovoltage and its unit (V, mV, uV, nV)	0, uV
amplifier gain	V is divided by this	1
invert voltage sign	Flips the sign of V	off
temperature column, temperature column is	Column of the temperature and what it holds (see above)	1, dT
T reference	Cold-side temperature for the hot-side mode	295
conductance column, unit, keep log G from / to	Optional conductance filter	empty
histogram of	`samples` or `file means`	samples
dT step	dT values are grouped to multiples of this	1 K
min. samples / files per dT	Fewer than this at a dT: that dT is left out	20
voltage bins, Seebeck bins	Histogram resolution	80, 80
S of the electrodes	Added to -dV/d(dT)	0 uV/K
bootstrap	Resamplings for the peak errors	100

The plots show the thermovoltage histograms at every dT with their Gaussians, the most probable V against dT with its standard errors (black) and widths (grey) and the fitted line with S, the S histogram, and the 2D histogram of S against log G.

On simulated traces with S = 12 and -8 uV/K the tab gives 11.9 +/- 0.4 and -8.1 +/- 0.4. Written out to files the way a real setup would (V in mV, the hot-side temperature, G in G_0) and read back with a conductance window, S = -6 comes back as -5.9 +/- 0.5.

When you report S, give its error, the dT values and how many junctions you measured at each, whether you used samples or file means, the conductance window, and whether you added an electrode Seebeck coefficient.

11. Piezo modulation tab

If a small sine is added to the piezo, the response of the conductance at that frequency gives the local decay constant:

$$\beta = -\frac{d \ln G}{dz}$$

For tunnelling β is about 2 κ, so around 20 per nm for gold in vacuum and less in solvents. On a molecular plateau it's small, because stretching the junction hardly changes G.

The modulation frequency is taken as the strongest narrow line in the piezo spectrum, unless you type it in. A line only counts if it's at least 20 times higher than the spectrum around it, so the normal piezo ramps aren't mistaken for a modulation. The recording is then split into windows of a few periods; in each window ln G and z are detrended and demodulated with a digital lock-in, and β is minus the in-phase ratio of the two amplitudes.

Setting	Meaning	Default
file	Which recording	first
from / to (s, big files)	For a TDMS file read in pieces (4.7): the part of the file to analyse; empty = all	empty
modulation frequency (Hz)	0 = find it automatically	0
periods per window	How long the lock-in averages	4
auto search above (Hz)	Lowest frequency the detection looks at	10
log G from / to	Only windows with mean log G in this range are shown	-6 / 1
displacement ratio	Junction nm per piezo nm (like in 4.5)	1

The top plots are the spectra of the piezo and of log G with the modulation frequency marked, the bottom ones β against log G for every window and the β histogram. Without modulation in the file you get "No piezo modulation was found".

12. Simulation tab

The simulation makes a fake recording where the molecular conductance, plateau length and noise exponent are known. I use it to get to know the settings, to pick cutting limits and noise settings, and to check an analysis does what it should before I trust it with real data.

Every simulated cycle goes through the same trace detection as real data and looks like this: approach until the contact snaps in at 1-4 G_0; pull through the atomic steps down to 1 G_0, break with snap-back, and then (with *molecule probability*) a molecular plateau, otherwise straight tunnelling decay; hold, where G fluctuates with 1/f noise whose power goes as G^n; retract into the noise floor; short hold at the end.

Setting	Meaning	Default
number of traces	Number of cycles	200
sampling rate, speed	Hz and junction speed in nm/s	10000, 2
displacement ratio	Junction nm per piezo nm for the piezo signal	1
tunnelling decay β	1/nm	10
contact size range, atomic plateau length, snap-back	Contact model	1-4 G_0, 0.08 nm, 0.3 nm
molecule probability	Fraction of traces with a molecular plateau	0.7
molecule log G, spread	Centre and spread (decades) of the molecular conductance	-3.5, 0.25
plateau length, SD, slope	Mean length and spread in nm, slope in decades/nm	0.4, 0.15, 0.3
2nd molecule log G, fraction	A second class, to test the clustering	off, 0
hold time, pull before hold, retract after hold	s, nm, nm	1, 1, 1
noise exponent n, flicker amplitude	The real n and how big the noise is	1.5, 0.05
noise floor log G, floor noise, white noise	Measurement floor and white noise	-5.5, 0.3, 0.02
bias, random seed	Bias in V; the same seed gives the same data	0.1, 1

Click Generate. The recording is added (or replaces what's loaded if Replace loaded data is ticked),

the traces are detected with the current settings and the status bar shows the true mean molecular conductance and n. Then check: the histogram peak should be at *molecule log G*, the mean plateau length close to *plateau length* (a little off, depending on the window), the noise exponent equal to *noise exponent* within its error, and with a *2nd molecule* set the clustering should separate the two.

13. Saving results and settings

File -> Save results (Ctrl+S) writes everything that has been calculated into one folder, plus the figures and a complete list of the settings that were used. You choose where the folder goes and what it's called; by default it's the name of the first recording plus `_Analysis`. If the folder is already there you're asked whether to overwrite it; No makes a new folder with _2, _3 and so on at the end. Every file name starts with the folder name, written `<name>` below.

The text files are tab-separated with one header line, so Origin, Excel and Python read them directly. The figures go into a `figures` subfolder (13.1).

File	What's in it	From
`<name>_config.json`	All settings of all tabs, the loaded files and the trace summary	always
`<name>_Good_Curves.txt`	All accepted traces: trace, file, z_nm, logG_G_0	Data & Traces
`<name>_all_traces.npz`	All traces including the rejected ones (NumPy file)	Data & Traces
`<name>_logHist.txt`	1D histogram, logG_G_0 and counts	Histograms
`<name>_3D_Histogram.txt`	2D histogram as a matrix, rows = log G, columns = distance	Histograms
`<name>_3D_hist_scales.txt`	Bin centres, distance_nm and logG_G_0	Histograms
`<name>_clusterN_logHist.txt`, `..._3D_Histogram.txt`, `..._3D_hist_scales.txt`	The same for each cluster	Histograms, if there are clusters
`<name>_peak_fit.txt`	Centre, error, G, σ, FWHM, amplitude and dwell time of each peak	Histograms, after a fit
`<name>_plateau length.txt`	Plateau length of every trace, one column per window	Plateau length
`<name>_Plateau_length_parameters.txt`	Statistics of each window and the tunnelling β	Plateau length
`<name>_Correlation.txt`, `_Correlation_scales.txt`	Correlation matrix; bin centres and mean histogram	Correlation
`<name>_Clustering_Results.txt`	Traces per cluster, silhouette, settings	Clustering
`<name>_cluster_labels.txt`	Trace, cluster, PC1-PC3	Clustering
`<name>_cluster_logHist.txt`	1D histogram of each cluster	Clustering
`<name>_Noise_windows.txt`	Every window: segment, start, samples, G, noise power, kurtosis, drift, ADF p, accepted	Flicker noise
`<name>_Noise_PSD.txt`	Frequency, mean PSD, floor PSD	Flicker noise
`<name>_Noise_scaling.txt`	n, Pearson r, Kendall τ, 2D Gaussian rho	Flicker noise
`<name>_Noise_bell_hist.txt`	Bell histogram as a matrix	Flicker noise
`<name>_Noise_summary.txt`	n with errors for all estimators, bell parameters, counts, settings	Flicker noise

File	What's in it	From
<name>_IV_DATA/IV_Curves_F.txt, IV_Curves_B.txt	V/I column pairs for every forward / backward curve	I-V
<name>_IV_models.txt	Per curve: low-bias G, V_t^+, V_t^-, ε_0 and a from TVS, ε_0, Γ and a from the fit with errors, R^2	I-V, after the model fit
<name>_IV_models_summary.txt	Fits of the mean curves, medians, settings	I-V, after the model fit
<name>_IV_most_probable.txt	Most probable current per bias bin, σ, points, fitted current	I-V, after the model fit
<name>_events.txt	Every event: segment, sample, time, mechanical yes/no, dG/dt, log G before and after	Events
<name>_events_per_trace.txt	Flickers and mechanical events per trace and its class	Events
<name>_hold_time_histogram.txt, ..._scales.txt, <name>_hold_most_probable.txt	Conductance-time histogram of the holds and the most probable log G per time	Events, with holds
<name>_events_summary.txt	Counts and settings	Events
<name>_EGaIn_<set>_logJ.txt	Per bias: $<$log|J|$>$, $\sigma_{\log}$, CI, sweeps	EGaIn
<name>_EGaIn_<set>_sweeps_logJ.txt	log|J| of every sweep on the bias grid	EGaIn
<name>_EGaIn_<set>_logR.txt, ..._Vtrans.txt, ..._summary.txt	Rectification per bias, V_{trans} per cycle, summary	EGaIn
<name>_EGaIn_beta.txt	β, J_0 and the points of the fit	EGaIn, after Fit β
<name>_Thermo_<set>_peaks.txt	Per dT: most probable V, σ, error, count	Thermopower
<name>_Thermo_<set>_seebeck_values.txt	S of every sample or file (and log G)	Thermopower
<name>_Thermo_<set>_voltage_histograms.txt, ..._summary.txt	Voltage histogram per dT; S, slope, settings	Thermopower
<name>_junctions.txt	Per trace: junction yes/no, flat plateau length, plateau log G	Junction statistics
<name>_junction_blocks.txt	Per block: mean trace number, yield, its error, median plateau log G, traces	Junction statistics
<name>_junction_summary.txt	Yield, medians, drift, settings	Junction statistics
<name>_ruptures.txt	Per trace: break position, Chow F, plateau log G, length to rupture, slopes, junction yes/no	Change points
<name>_step_levels.txt, <name>_step_level_histogram.txt	Every PELT level (log G, length, slope, plateau yes/no) and the length-weighted level histogram	Change points
<name>_changepoints_summary.txt	Fractions, medians, settings	Change points
<name>_blinks.txt	Every blink: segment, start, duration, log G, baseline, jump height	Blinking
<name>_blink_off_times.txt, <name>_blink_histogram.txt	Times between blinks; histogram of the blink conductance	Blinking

File	What's in it	From
<name>_blinking_summary.txt	Blink conductance, lifetimes, rate, settings	Blinking
<name>_piezo_modulation.txt	t, logG, amplitudes, β and phase per window	Piezo modulation
figures/<name>_<plot>.pdf and .png	Every plot as its own figure (13.1)	every tab
<name>.opju	Origin project with the plot data and graphs, if "plot with" includes Origin (13.1)	every tab

Plateau length, Junction statistics and Correlation results get the cluster name added (e.g. <name>_cluster2_Correlation.txt) if they were calculated for one cluster.

13.1 Figures

Every plot is saved as its own figure, sized for a journal. Nothing is combined into one image, except that 2D histograms keep their colour bar, because you can't read them without it. Multi-panel figures can be put together afterwards in Illustrator, Inkscape or similar.

	Default	Nature asks for
Width	89 mm (one column)	89 mm single, 120 mm 1.5 columns, 183 mm double
Font	Arial 7 pt, tick labels and legends 6 pt	sans-serif, 5-7 pt
Lines	data 0.75 pt, axes and ticks 0.5 pt	at least 0.5 pt
Colours	Okabe-Ito palette (colour-blind safe), viridis for maps	colour-blind friendly
Titles	none, the file name says what it is	no panel titles
Files	PDF with editable text, plus a 600 dpi PNG	vector, or at least 300 dpi

All of this can be changed under File -> Figure export settings (width, font, font size, line widths, tick direction, formats pdf/png/tiff/svg/eps, resolution). The settings are remembered and also written to *_config.json.

Figure files (figures/<name>_...)	What they show	Tab
example_traces	Random accepted traces	Data & Traces
logG_histogram, 2D_histogram, clusterN_logG_histogram, clusterN_2D_histogram	1D and 2D histograms of all traces and of each cluster	Histograms
plateau_length_histogram, tunnelling_decay_histogram	Plateau lengths and tunnelling β	Plateau length
correlation_map, correlation_mean_histogram	Correlation map and mean histogram	Correlation
clustering_PCA_scree, clustering_PCA_scatter, clustering_logG_histograms, clustering_silhouette_scan, clustering_clusterN_2D_histogram	PCA and cluster results	Clustering
noise_example_segment, noise_PSD, noise_scaling_fits, noise_bell_histogram, noise_correlation_vs_n, noise_fit_residuals	Flicker noise	Flicker noise

Figure files (figures/<name>_...)	What they show	Tab
junction_plateau_conductance_histogram, junction_conductance_vs_trace, junction_yield_vs_trace, junction_plateau_length_histogram, junction_length_vs_conductance	Junction yield and plateau statistics	Junction statistics
IV_curves, IV_current_histogram, IV_conductance_histogram, IV_dIdV_histogram	I-V curves and histograms	I-V
events_example_trace, events_example_derivative, events_trace_classes, events_flickers_per_trace, events_transition_map, events_hold_time_histogram	Events	Events
changepoints_example, changepoints_rupture_conductance, changepoints_chow_F, changepoints_rupture_length, changepoints_levels, changepoints_step_heights	Rupture and step analysis	Change points
blinking_example, blinking_conductance_histogram, blinking_on_times, blinking_off_times, blinking_level_vs_time, blinking_aligned_histogram	Blinking	Blinking
EGaIn_<set>_egain_JV, ..._logJ_2D_histogram, ..._logJ_histogram, ..._rectification_histogram, ..._transition_voltage_histogram, ..._SLM_fit, ..._beta	Ensemble junctions, one set per data set	EGaIn
Thermo_<set>_thermo_voltage_histograms, ..._thermo_voltage_vs_dT, ..._thermo_seebeck_histogram, ..._thermo_seebeck_vs_logG	Thermopower, one set per data set	Thermopower
IV_SLM_fit, IV_Fowler_Nordheim, IV_transition_voltage_histogram, IV_SLM_parameters, IV_eps0_histogram	Single-level model and TVS	I-V, after the model fit
modulation_piezo_spectrum, modulation_logG_spectrum, modulation_beta_vs_G, modulation_beta_histogram	Piezo modulation	Piezo modulation

The plots on screen are drawn by the same code, so the exported figures look the same, just at journal size. If you want the whole screen view as one picture, use the save button in the plot toolbar.

Plotting in Origin instead. If you finish figures in Origin, set "plot with" in File -> Figure export settings to `Origin` (or `matplotlib + Origin` for both). Save results then also puts every plot into Origin: the plotted data go into a workbook (heatmaps into a matrix book) and each plot is rebuilt as an Origin graph with the same page size, axis box, fonts, line widths, colours, axis ranges, legend and labels as the exported figure, so they look the same and stay editable. Histogram steps keep the original bin centres and counts next to the step outline, and heatmaps use the same colour map. Every tab gets its own folder in the Project Explorer, and the project is saved as `<name>.opju` in the results folder. With `Origin` alone no figure files are written. File -> Send plots to Origin does the same for the current plots without saving anything, which is handy for trying settings.

This needs Windows, Origin 2021 or newer and the Python package originpro (`py -m pip install originpro`). If Origin is already open the graphs go into the open project, otherwise Origin is started; either way it stays open. Origin is controlled from the program, so leave it alone until the status bar says it's done (about 2 s per plot). If Origin has a dialog open, for example a licence reminder, close it first.

13.2 Settings and converted files

File -> Save settings writes all settings to a JSON file and File -> Load settings reads them back; a `*_config.json` from a results folder works too, which is handy for repeating an analysis. The `.cvr` (and `.csv`) files from the Convert tab are written separately, see 3.2.

14. Troubleshooting

Problem	What to do
All traces rejected with "never reaches high limit"	The contact doesn't get above *high limit to check*. Lower it to just under the contact level you see in the overview. If the amplifier saturates below 1 G_0, check *Ia saturation* so saturated points count as contact (3.3).
All traces rejected with "never reaches low limit"	The noise floor is above *low limit to check*. Put both low limits about half a decade above the floor. With holds in the measurement, leave *join pull-hold-pull* on.
Conductance off by a constant factor	For `.cvr` files the *constant bias* in the column settings has to match the measurement; for other files check the *Ia / Ib scale* and *conductance unit*. For TDMS, look at the bias channel and series resistor in the Convert preview.
Distance axis is noisy or zig-zags	More *piezo smoothing*, or *distance from* `index` with the right piezo rate.
Distances far too long or short (MCBJ)	Calibrate the *displacement ratio* with the tunnelling fit (6.3).
"No hold segments were found"	No piezo stops in the recording, or no piezo signal at all. Use *segments* = `plateau windows`, or measure with holds.
"Only N accepted windows"	Widen *accept log G*, make *peak sigma* or *n sigma* bigger, or switch drift, kurtosis and ADF off one at a time to see which one throws the windows out.
"The integration band contains fewer than 2 frequency points"	The windows are too short for the band. Make *window* longer or raise *integrate from*.
OLS and Theil-Sen give different n	There are outliers. Use Theil-Sen, look at the residual plot, maybe switch on the ADF test.
"No piezo modulation was found"	There's no modulation in the file or it's too weak. Type in the *modulation frequency* if you know it.
"No bias sweeps were found"	The bias is constant; I-V needs sweeps.
No transition voltage (empty V_t)	The maximum of $V^2/\lvert I\rvert$ is at the end of the sweep, so the bias didn't reach V_t. Sweep further if the junction survives it, or rely on the model fit.
SLM errors huge, ε_0 and Γ on a diagonal streak	The bias range is far below ε_0, so the curves can't separate ε_0 and Γ. Report the low-bias conductance instead, or measure to higher bias.
Events: nothing found although steps are visible	Check *derivative dt* and the thresholds: they are on the dG/dt scale with G in G_0. Look at the derivative plot and put the thresholds under the peaks you want. With *peak detection* = `minimum width`, lower *peak width*.
Events: very many flickers on noisy data	Raise the flicker threshold or the low-pass strength; check *valid from / to log G* so the noise floor is excluded.
EGaIn: every junction is a short	*current unit* or *contact diameter* is wrong (J too large), or the short threshold is too low.
EGaIn column pairs: fewer junctions than pairs	Pairs with fewer than 4 valid points are skipped; a lone last column without a partner is ignored.

Problem	What to do
Thermopower: "fewer than ... at every dT"	Lower *min. samples / files per dT*, or check *dT step* and the temperature column mode (dT vs hot-side temperature).
Thermopower: S has the wrong sign for a known molecule	Tick *invert voltage sign*; the sign of the measured voltage depends on which electrode is grounded.
EGaIn: no rectification or V_{trans}	The sweeps don't cover both polarities in one cycle, or the bias doesn't go far enough for a V_{trans}.
Rashid noise preset accepts almost nothing	The holds start or end off the molecular plateau (+/- 1 σ is strict). Check that *peak centre* is set, or relax *ends within +/- n sigma*.
Junction yield close to 100 % for a blank measurement	Tunnelling counts as plateau: lower *flat if slope below* or raise *junction if flat length above* (6.4).
Junction yield close to 0 % although there's a clear peak	The plateaus are tilted more than *flat if slope below*, or the window misses them; use Use peak from histogram fit.
Clustering changes from run to run	Keep the *random seed* fixed (it's 0 by default).
A big TDMS file makes the program slow or run out of memory	Lower *read in pieces above* (4.7) so the file is read in pieces. For I-V and modulation, give a *from / to* range instead of the whole file.
"Detect / cut traces" fails for a file read in pieces	The file was moved or renamed after loading; it is read again for every detection. Load it again from its new place.
Window empty or tiny on a 4K screen	Set the Windows display scaling to 150-200 %.
"No column ... available: ..."	The column name or number isn't in the file. The message lists what is there; Show columns (4.1) does too. Numbers start at 0.
Igor file: "No numeric waves found"	The file has only text waves, or igor2 is missing (`py -m pip install igor2`).
Excel, MATLAB v7.3 or HDF5 file: "... needs openpyxl / h5py"	Install the missing package (`py -m pip install openpyxl h5py`); the exe has both.
A text file with a comment line at the top isn't read	Comment lines are skipped as long as the data lines start with a number. If the header has a different number of names than there are columns, the columns are just numbered (Show columns).
Change points: almost every trace is a junction, also for a blank measurement	Lower *max plateau slope* or raise *Chow F threshold*; look at the Chow F plot (6.5).
Change points: PELT gives many levels per trace	Raise *PELT penalty factor* or *noise at least*.
Blinking: hundreds of very short blinks	The up threshold is in the noise. Raise it, or smooth more.
Blinking: no blinks although the jumps are visible	The blink level is outside *blink log G from / to*, or the molecule is in the gap more than 80 % of the time (lower the baseline percentile).
Error message about a file	The file is broken or isn't break-junction data. Open it in a text editor or a TDMS viewer and have a look.

You can load several files at once; they're analysed together, the trace numbers just continue and every trace remembers which file it came from. The program never changes your data files, it only reads them and writes results to the folder you pick.

To repeat an old analysis, load the `*_config.json` from its results folder with File -> Load settings, load the same data and run the same tabs.

To check that an installation works, open a command prompt in the program folder and run `GZero.exe -selftest report.txt` (or `py GZero.py -selftest report.txt`). It runs the simulation, trace detection, histogram fit, dwell time, clustering, noise analysis, junction yield, event detection, I-V detection, the single-level fit (also at 300 K on the energy grid with Levenberg-Marquardt) and TVS, the EGaIn statistics with β on simulated junctions and a column-pair file, the thermopower fit, reading

a large TDMS file in pieces, every file format (gzip text, Excel, MATLAB, NumPy, HDF5, LabVIEW), blink detection, change-point detection and the figure export without opening a window and writes the results to `report.txt`. If everything is fine the last line is `SELFTEST OK`.

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Entries 1-24 were checked against Crossref on 8 October 2026. The table at the end lists what to cite for which analysis.

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37. J. R. Widawsky, P. Darancet, J. B. Neaton, L. Venkataraman, "Simultaneous determination of conductance and thermopower of single molecule junctions", *Nano Letters* 12, 354-358 (2012). doi:10.1021/nl203634m Conductance and thermopower in the same STM break-junction measurement; sign of S for HOMO- and LUMO-conducting junctions.

15.7 Change points and blinking

38. J. M. Hamill, W. Bro-Jorgensen, Z. Balogh, H. Li, S. Leitherer, D. Solomon, A. Halbritter, G. C. Solomon, "Improving single-molecule conductance measurements with change point detection from the econometrics toolbox", arXiv:2401.12769 (2024). arxiv.org/abs/2401.12769 Chow test for the rupture of each trace; plateau conductance per junction. Journal version not checked.

39. G. C. Chow, "Tests of equality between sets of coefficients in two linear regressions", *Econometrica* 28, 591-605 (1960). doi:10.2307/1910133
40. R. Killick, P. Fearnhead, I. A. Eckley, "Optimal detection of changepoints with a linear computational cost", *J. Am. Stat. Assoc.* 107, 1590-1598 (2012). doi:10.1080/01621459.2012.737745 The PELT algorithm.
41. A. Vaidya, J. Dyer, O. L. A. Monti, "Understanding conductance fluctuations in single-molecule break junctions: a statistical approach", APS March Meeting 2026, abstract. meetings-archive.aps.org PELT-R: a variable number of change points per trace.
42. W. Haiss, H. van Zalinge, S. J. Higgins, D. Bethell, H. Hobenreich, D. J. Schiffrin, R. J. Nichols, "Redox state dependence of single molecule conductivity", *J. Am. Chem. Soc.* 125, 15294-15295 (2003). doi:10.1021/ja038214e The I(t) (blinking) method.
43. W. Haiss, C. Wang, I. Grace, A. S. Batsanov, D. J. Schiffrin, S. J. Higgins, M. R. Bryce, C. J. Lambert, R. J. Nichols, "Precision control of single-molecule electrical junctions", *Nature Materials* 5, 995-1002 (2006). doi:10.1038/nmat1781 I(s) and I(t) measurements at controlled electrode spacing.

The details of 26-43 were taken from the publishers' and university repositories' pages, arXiv and the code repositories in October 2026. They haven't been re-checked against Crossref like 1-24.

15.8 What to cite for each analysis

Analysis in this program	Cite
Conductance histograms, plateau length	2, 3 (technique); 4 or 5 (review)
Junction yield, plateau conductance per trace	26
2D cross-correlation	11
Clustering	12, 13; 15 or 14 for related methods
Flicker noise, bell / 2D Gaussian	7
Flicker noise, OLS / zero correlation	8
Flicker noise, ADF + Theil-Sen	10, with 17, 18, 20
Single-level model fit of I-V curves	27
Transition voltage spectroscopy	28, 29
Single-level fit at finite temperature, most probable I-V, Rashid noise preset, events, dwell time	30 (31 for the integration)
EGaIn statistics, yield, rectification	32; 34, 35 for the software you compare with
β and J_0 from a length series	33
Thermopower, Seebeck coefficient	36, 37; 34 for the XMe thermopower module
Rupture of each trace, plateau conductance per junction (Change points)	38, 39
Steps between levels (PELT)	40, 41
Blinking / I(t)	42, 43
PSD (Welch), smoothing	22, 23
Libraries used	24, 25